SECTION 2 · SUBSECTION 2.1
SOFTWARE LOAD, SD CARD ROUTE

2.1.1 INTRODUCTION.

InkOS loads onto the X4 with an iPhone only: the cloud builds the image, the phone carries it to the microSD card, and the bootloader is reported to write it. The file must be named update.bin, in the card root (figure 2.1-1).

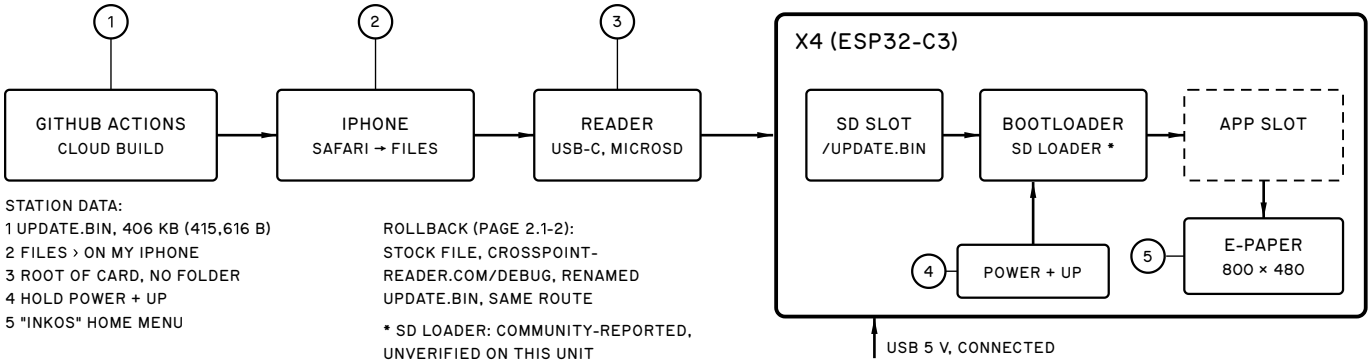


Figure 2.1-1. Software Load Path, Cloud to App Slot IKS-501A

2.1.2 FUNCTIONAL DESCRIPTION.

A passing build posts update.bin to the release LATEST. Prove the route first with the CrossPoint release file, then load InkOS the same way; any failure goes to page 2.1-2. On first boot, SETTINGS › DIAGNOSTICS shows each raw reading (figure 2.1-2).

2.1.2.1 Button Decode Bands.

Six buttons share two resistor ladders on GPIO 1 and 2. A reading outside its band is fixed in BOARD.H, not in hardware.

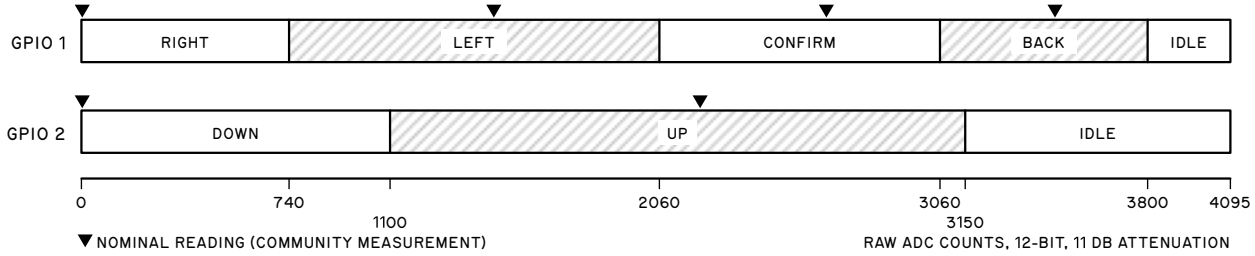


Figure 2.1-2. Button Decode Bands, Firmware v0.1 IKS-502A

2.1.3 LOAD DATA.

Table 2.1-1. InkOS v0.1 Load Data

Parameter	Value
Image file	UPDATE.BIN, 406 KB
Partition used	30.1 % of 1.25 MB build
RAM at build	62 KB of 320 KB
Load keys	POWER + UP, USB on
Display	800 × 480, SSD1677 assumed
Button poll	5 ms / 20 ms idle

CAUTION

Keep USB power on until the load ends; a cut can corrupt the app slot.

```
LOAD - INKOS                                  OPR
1  FILES - UPDATE.BIN TO CARD ROOT
   √ SIZE - 406 KB, NO "-1" IN NAME
2  CARD IN, USB ON - POWER + UP HOLD
   √ HOME MENU - "INKOS"
3  DIAGNOSTICS - PRESS ALL 6
   √ COUNTS - IN BAND (FIG 2.1-2)
```

SECTION 2 · SUBSECTION 2.1 (CONTINUED)
SOFTWARE LOAD, MALFUNCTION AND ROLLBACK

2.1.4 PREREQUISITES.

A microSD card formatted FAT32, a USB-C card reader for the iPhone, a USB charger and the X4 charged. No full-flash backup can be made from an iPhone; the stock file in 2.1.6 is the only restore point, so keep a copy in Files before the first load.

2.1.5 LOAD DECISION SEQUENCE.

The proof load separates the route from the code: if the CrossPoint release file will not load, InkOS will not either, and the fault is the unit or the route (figure 2.1-3).

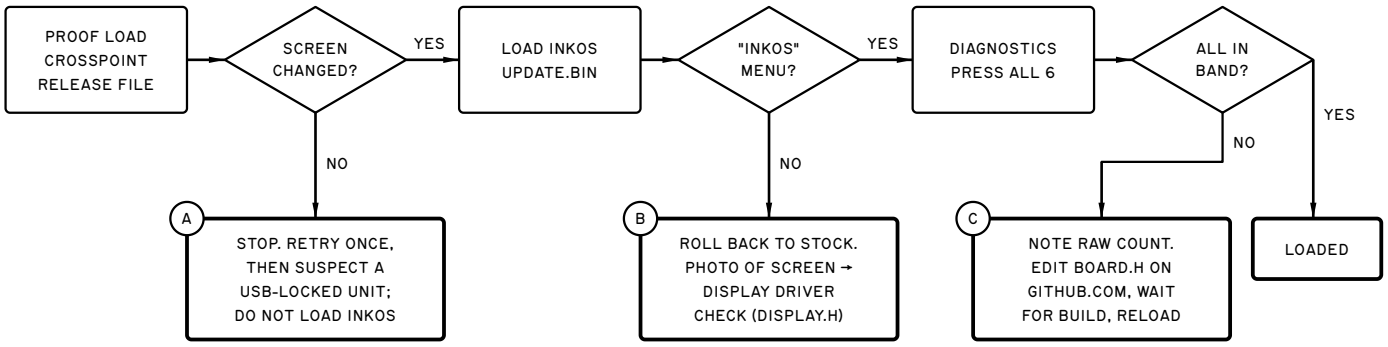


Figure 2.1-3. Load Decision Sequence and Malfunction Branches IKS-503A

2.1.5.1 Unverified Elements.

The SD loader, the load keys and the SSD1677 controller come from community reports, not from this unit. Treat the first load as a test, not a routine.

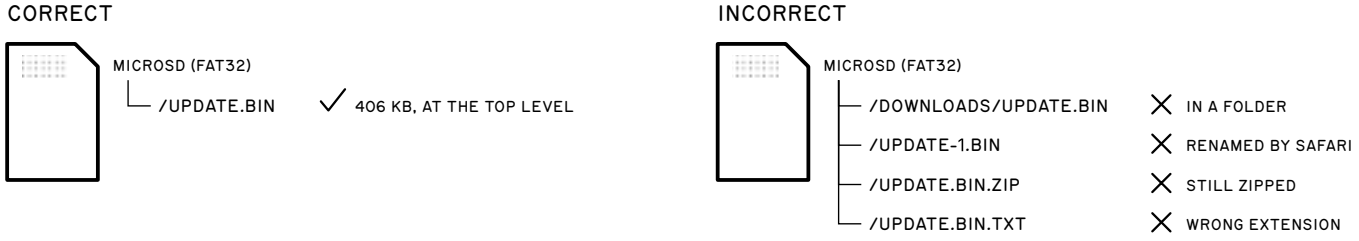


Figure 2.1-4. Card Root, Correct and Incorrect Placement IKS-504A

2.1.6 MALFUNCTION DATA.

Table 2.1-2. Load Malfunctions

Symptom	Action
No reaction to POWER + UP	Check name, root, FAT32, USB; retry once (branch A)
Name ends in -1	Delete old; rename (fig 2.1-4)
Blank screen	Roll back; photo (branch B)
Button misreads	Note the raw count (branch C)
Clock shows ~	Power was lost; set date and time
GitHub run failed	LATEST holds the prior build

N O T E

POWER is not shown in DIAGNOSTICS. Verify it apart: a short press forces one full refresh.

ROLLBACK - STOCK **OPR**

```
1  SAFARI - CROSSPOINTREADER.COM/DEBUG
2  STOCK FIRMWARE - DOWNLOAD
3  FILES - RENAME UPDATE.BIN
4  FILES - COPY TO CARD ROOT
5  CARD IN, USB ON - POWER + UP HOLD
   √ STOCK READER SCREEN
```

LOAD